

PARTS LIST

Semiconductors:

T1	- SS9014 npn transistor
LED1	- 5mm blue LED
D1	- 1N4007 rectifier diode
ZD1	- 5V, 1W Zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2	- 47-ohm
R3	- 470-kilo-ohm
R4	- 10-kilo-ohm

Capacitors:

C1	- 100 μ F, 63V electrolytic
C2	- 0.47 μ F, 400V(CBB22) polyester

Miscellaneous:

CON1	- 2-pin female Berg connector
LDR1	- 5mm LDR (GL5528)

(All these are available from kitsnspares.com)

difficult. Even though the bill-of-material (BOM) includes leaded components to save space (height) and improve mechanical stability, it is preferable to use lead-less (SMD) components if you are an advanced solder artist.

After assembling the components, enclose the circuit board in a suitable waterproof and flame-retardant box. The enclosure design depends on

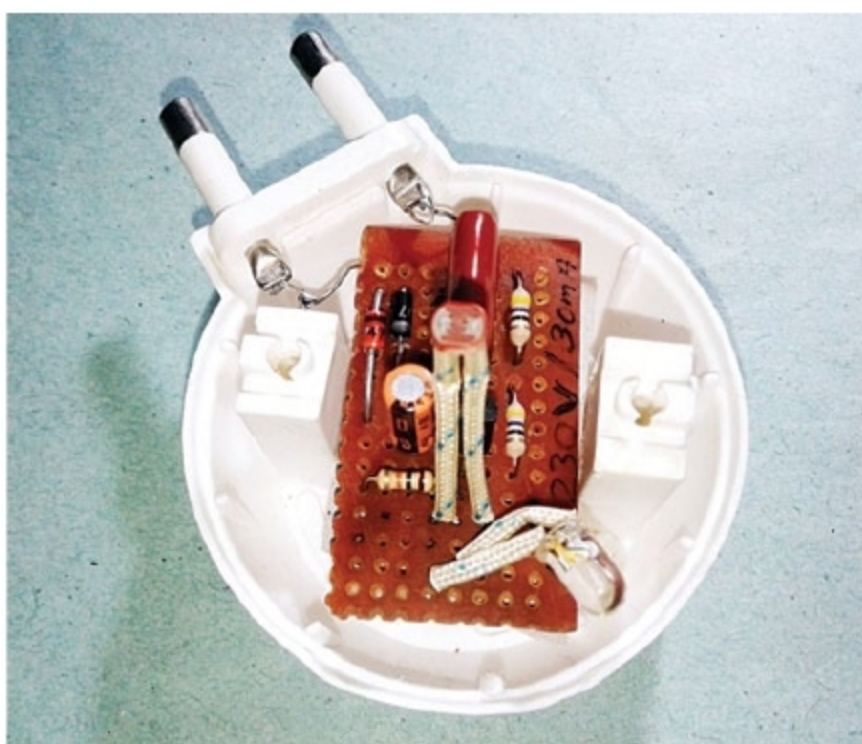


Fig. 6: Author's prototype

your artistic background and skill, which makes DIY construction an attractive experience. The author's prototype wired on a perf board and enclosed in a defunct liquid vaporiser is shown in Fig. 6.

Note. Capacitive power supplies are not safe due to lack of isolation.

But in sealed-off devices these can be a great way of getting the working voltage for certain applications. Obviously, such devices have power factor (PF) close to zero, making it questionable whether these meet the mandated PF laws.

Caution. There are chances of electric shock during experimentation with a transformer-less circuit interfaced to grid power.

So, you must be very careful while handling this circuit. **EFY**



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